Curriculum Vitae

Mostafa Sameer

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Academic Background

Ph.D. 2024–Present	**Civil Engineering**, Florida International University, Miami, FL Advisor: Dr. Wallied Orabi, Associate Professor, Moss School of Construction GPA: 3.90/4.00 Research: Interpretable bridge rehabilitation prioritization integrating structural condition, transportation network efficiency, and community accessibility.
M.S. 2023–2024	**Civil Engineering, Transportation Engineering**, Wayne State University, Detroit, MI Advisor: Dr. Xiaodong Qian, Assistant Professor, Civil and Environmental Engineering GPA: 3.47/4.00
Graduate Study 2023–2024	**M.S. Civil Engineering, Transportation Engineering**, King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia Transferred to Wayne State University to continue graduate study Advisor: Dr. Hassan M. Al-Ahmadi CGPA: 3.00/4.00
M.B.A. 2017–2019	**Operations and Supply Chain Management**, University of Chittagong, Bangladesh Advisor: Dr. Sagib Kumar Ghosh CGPA: 2.99/4.00
B.Sc. 2011–2016	**Civil Engineering**, Ahsanullah University of Science and Technology, Dhaka, Bangladesh Thesis: Determination of Trip Attraction Rates of Different Size Hospitals in Dhaka City

Professional Experience

Graduate Research Assistant 2026–Present	**Florida International University**, Moss School of Construction, Miami, FL • Developing GIS-based coastal exposure and risk workflows to quantify resilience across multiple spatial resolutions. • Developing an interpretable bridge rehabilitation prioritization framework using structural condition, transportation network efficiency, and community accessibility.
Graduate Research Assistant 2024–2025	**Digital Twin Lab, Florida International University**, Miami, FL • Investigated hurricane-induced wind-driven rain intrusion through sliding glass door systems in partnership with the Florida Building Commission. • Co-authored the technical report, *Standards Review for Wind-Driven Rain Intrusion through Sliding Glass Door Tracks* (DOI: 10.13140/RG.2.2.21522.16322).
Quality Assurance Engineer / Technician 2024	**Fluor Corporation**, Gordie Howe International Bridge, Detroit, MI • Performed QA/QC procedures for structural elements in accordance with ASTM and AASHTO standards on a USD 5.6 billion binational bridge megaproject. • Managed compliance documentation, concrete slump testing, air-density/pressure testing, and field coordination with testing stakeholders.
Graduate Teaching Assistant 2023–2024	**Emerging Mobility Lab, Wayne State University**, Detroit, MI • Developed Python-driven simulation experiments in CARLA/Unreal Engine 4 to evaluate traffic optimization strategies. • Modeled V2X assumptions for e-scooters and mixed traffic networks and assessed efficiency and safety trade-offs.

Professional Experience (continued)

Graduate Teaching Assistant
2022–2023

King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia
- Taught more than 85 undergraduate students in Drafting and Transportation Engineering; designed network models using Synchro, VISSIM, and AutoTURN.

Civil Engineer
2019–2022

KDS Textile Mills Ltd., Chattogram, Bangladesh
- Managed factory load distribution and structural considerations to support building safety and structural integrity.
- Applied AutoCAD, ETABS, STAAD.Pro, GIS, and Civil 3D for analysis, design, and project support; supervised industrial construction activities.

Senior Officer, Civil Construction
2017–2019

Youngone Sports Shoes Industries Ltd., Chattogram, Bangladesh
- Monitored factory building structural integrity, supported safety compliance, and prepared structural condition reports identifying corrective actions.

Assistant Engineer
2016–2017

Luminars Engineers Association, Chattogram, Bangladesh
- Conducted structural and building inspections, evaluated existing conditions, and documented engineering findings supporting retrofit and rehabilitation.
- Assisted in evaluating the structural integrity of existing buildings and documenting engineering findings.

Technical Skills

Programming: Python, R, MATLAB
Simulation: CARLA, Unreal Engine 4, SUMO, AIMSUN, VISSIM, Synchro, AutoTURN
GIS and Design: ArcGIS, QGIS, AutoCAD, Civil 3D
Structural Analysis: ETABS, STAAD.Pro

Honors and Awards

- Florida Professional Engineers in Government (FPEG) Scholarship (2026).
- Florida Water Environment Association (FWEA) FE Sponsorship (2026).
- Barnes & Noble Graduate Scholarship, Florida International University (2025, 2026).
- First Prize, Flood Guard Miami, EmpowHer MiniHack (2025).
- Sigma Xi Research Honor Society, Member (2025).
- Graduate Writing Mentorship Fellow, FIU (2025).
- Academic Merit Scholarship, King Fahd University of Petroleum and Minerals (2022–2023).
- Licensed Engineer, Institute of Engineers Bangladesh (2022).

Leadership and Service

- RSO Representative, Florida Engineering Society (FES), 2024–Present.
- Secretary, Bangladesh Student Organization (BSO), FIU, 2025–2026.
- RSO Representative, Florida Water Environment Association (FWEA), 2026–Present.
- Active Graduate Member, ASCE FIU Student Chapter, 2024–2025.
- Active Member, ASCE WSU Student Chapter, 2023–2024.
- Facilitator, Global Talent Accelerator, Global Detroit (2023).
- Volunteer: NHERI/Natural Hazard Engineering Research Webinar and Technical Activities, FIU Research, 2025–Present; Michigander Scholars Exchange Program (2024).

Publications

Journal and Conference Publications

- Gazder, U., Islam, M. K., & Sameer, M. (2026). Long-Term Transportation Fuel Consumption in Annex I Countries: A Mixed-Method Multi-Sector Analysis (1990–2020). *Energy Engineering*. https://doi.org/10.32604/ee.2026.086249
- Aldossari, A., Alhajji, B., Hanbzaza, R., Haque, M. E., Islam, M. K., & Sameer, M. (2026). Data-Driven Decarbonization: Clustering of National CO_2 Emission Profiles and Economic Indicators for Sustainable Energy Transitions. *Energy Engineering*. https://doi.org/10.32604/ee.2026.088402
- Islam, M. K., Hassan, M., Sameer, M., & Gazder, U. (2026). Energy and Emission Implications of Electric and Autonomous Vehicle Deployment in Saudi Arabia's Urban Transport Systems. https://doi.org/10.1109/FET68771.2026.11601878
- Islam, M. K., & Sameer, M. (2026). Advancing Sustainable Mobility in GCC Countries: A Pathway to Achieving Sustainable Development Goals. *Proceedings of the 3rd International Conference on Sustainability*. https://doi.org/10.1007/978-981-92-1726-7_17
- Hassan, M., Kabir, M. E., Islam, M. K., Rambe, A. H., Jusoh, M., & Sameer, M. (2025). Mapping the Machine Learning Landscape in Autonomous Vehicles: A Scientometric Review of Research Trends, Applications, Challenges, and Future Directions. *IEEE Access, 13*, 182036–182077. https://doi.org/10.1109/ACCESS.2025.3620637
- Zaman, T. A. B., Haque, T., Sen Gupta, H., Sameer, M., & Nofal, O. (2025). Computational Review of Disaster Resilience Research in Bangladesh. *Progress in Disaster Science*. https://doi.org/10.1016/j.pdisas.2025.100486
- Mahin, H. D., Hassan, M., An, H. K., & Sameer, M. (2025). Socioeconomic Mobility and Urban Travel Patterns in Post-COVID-19 United States: Insights from the 2022 National Household Travel Survey. https://doi.org/10.2139/ssrn.5192486
- Sameer, M., Soncco Sinchi, K. W., & Amuda, K. A. (2026). Human-Centered Data Fusion and Multi-Scale Digital Twins for Community Resilience Under Multi-Hazard Events. Preprint.

Technical Reports and Conference Contributions

- Impact of Modeling Resolution on Flood Risk Analysis Outcomes (2025). EMI 2025. Sameer, M., & Nofal, O.
- Ground Investigation and Response of Jhilmil Residential Town Project (2018). ICACE, CUET. Javed, S. A., Sameer, M., et al.
- Liquefaction Potentiality Analysis of Jhilmil Residential Town Project (2016). BUET–ANWAR ISPAT Civil Engineering Summit. Javed, S. A., & Sameer, M.

Research Frameworks and Technical Development

- NBI FHWA Bridge Research Analytics Scoring.
- Florida Statewide Bridge Network, Live Platform.
- Selected to advance to the full-submission stage, ASCE Global Innovate Competition: AI & Generative Models Challenge for Civil and Construction Engineering (2026).